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14 June 1991
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30 September 1980

MILITARY SPECIFICATION

CONTROL ASSEMBLY R.F. (UNIT 1) 11748205

This specification is approved for use by U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Control Assembly R.F. (Unit 1), hereinafter referred to as control assembly, used with the System Tester, Radar AN/VPM-2, 11748124 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-S-48431	-	Solid State X-Band Source 10548182
MIL-P-48606	-	Power Supply Assembly (Unit 2) 11748133

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1240

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STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-2000 - Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10549058 - Radar Set AN/VPS-2
11748133 - Power Supply Assembly (Unit 2)
11748144 - W5 Waveguide Flexible
11748155 - RF Amplifiers
11748168 - Microwave Integrated Assembly #l
11748205 - Control Assembly R.F. (Unit 1)
11748213 - Cable, W1 Power Supply
11748214 - Cable, W2 Signal Input
11748215 - Cable, W3 Power Input
11748216 - Cable, W4 Data Input
11748455 - Microwave Integrated Assembly #5
11748460 - Wire List

PACKAGING DATA SHEET

PS11748205 - Packaging of Control Assembly R.F. (Unit 1): 11748205

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 11748205, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the control assembly shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.3 Environmental service conditions.

3.3.1 Vibration. The control assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations; in each of three mutually perpendicular axes, as follows:

Frequency Range	AMplitude
10 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 55 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ±10% per decade.
The 10-55-10 hertz sweep time is 3 minutes 45 seconds ±10%.

3.3.2 Altitude.

3.3.2.1 Operating. The control assembly shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.2.2 Non-operating. The control assembly shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.3 Temperature.

3.3.3.1 Operating. The control assembly shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 degrees F and +155 degrees F.

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3.3.3.2 Non-operating (storage). The control assembly shall not suffer permanent damage when subjected to the constant temperature extremes of -60 degrees F for a period of 24 hours and +160 degrees F for a period of 4 hours (after temperature stabilization) and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.4 Humidity (non-operating). The control assembly shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.3.5 Immersion (non-operating). The control assembly shall meet the requirements of 3.4 and 3.5 and shall have no evidence of water penetration within the case after being imersed in water and subjected to a positive pressure equivalent to 3 head-feet of water (approximately 1.3 psi) for a period of two hours. The power supply assembly shall be immersed with its case closed, an equivalent dummy load, and a sealed plug in place of the pressure release valve.

3.4 Electrical. Each conductor shall be electrically continuous as specified on drawing 11748460.

3.5 Performance.

3.5.1 Primary input power. The control assembly shall be capable of meeting all requirements of this specification with +24 ± 6 Vdc primary input power applied to a Power Supply Assembly (Unit 2) P/N 11748133 that meets all the requirements of MIL-P-48606 or when operated from individual power supplies meeting the performance requirements of MIL-P-48606.

3.5.2 Warm-Up. An initial warm up period of 10 to 15 minutes shall occur prior to the commencement of any performance testing unless otherwise specified.

3.5.3 Definitions.

3.5.3.1 CW. The term “CW” shall define the condition of the control assembly in which RF switch “S1” in the microwave integrated assembly #1 (11748168) and the RF “gate switch” in the microwave integrated assembly #5 (11748455) are in a steady state low insertion condition. The control assembly shall be in the CW condition when cable W2 is disconnected from SIGNAL INPUT J3.

3.5.3.2 Pulse. The term “Pulse” shall define the condition of the control assembly in which RF switch “S1” in the microwave integrated assembly #1 (11748168) and the RF gate switch in the microwave integrated assembly #5 (11748455) are being controlled by their respective pulse drivers. The control assembly shall be in the Pulse condition when cable W2 is conncted to SIGNAL INPUT J3.

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3.5.4 Voltage monitor. The following requirements shall be met with POWER SUPPLY cable W1 connected to control assembly comector J2 and POWER INPUT Cable W3 connected to control assembly connector J4 .

a. With the FUNCTION SELECT switch in the VOLTAGE position and VOLTAGE SELECT switch in the positions delineated in Table I, the digital display (DVM) shall indicate the applied voltage specified in Table I within a tolerance of ± 0.050 Vdc.

TABLE I. Voltage monitor readings

VOLTAGE SELECT SWITCH POSITION	APPLIED VOLTAGE VALUE (VDC)	APPLIED
+24	+24 ± 0.200	W3-A(+) W3-B(ret.)
+5	+5 ± 0.200	W1-A(+) W1-B(ret.)
+15	+15 ± 0.300	W1-E(+) W1-F(ret.)
-15	-15 ± 0.300	W1-G(-) W1-F(ret.)
+28	+28 ± 0.600	W1-C(+) W1-D(ret.)

b. The voltage at cable end W1 between W1-H (ret.) and W1-1(+) shall be as defined in 3.5.4a for the VOLTAGE SELECT switch in the +24 position.

3.5.5 Voltage input monitor. With the FUNCTION SELECT switch in the VOLTAGE position and the VOLTAGE SELECT switch in the VOLT TP position, the Digital Display (DVM) shall indicate the applied inputs specified in Table II when connected to the VOLT TP input jack (Applied Input - Input, Ret - GND). NOTE: DO NOT EXCEED 36 VOLTS.

TABLE II. Voltage input monitor readings

APPLIED INPUT - VDC	DIGITAL DISPLAY (DVM)
+35.00 ± 00.10	+35.00 ± 00.15
Short	0.000 ± 0.010
Open	0.000 ± 0.015

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TABLE II. Voltage input monitor readings

APPLIED INPUT - VDC	DIGITAL DISPLAY (DVM)
-35.00 ± 00.10	-35.00 ± 00.15
+00.10 ± 00.01	+00.10 ± 00.02

3.5.6 STALO output.

a. All requirements shall be measured at the STALO OUT port with the MODE SELECT switch in the RADAR position unless otherwise indicated.

b. The control assembly shall be in the Pulse condition.

c. The RF output power shall be +12 dBm minimum for all positions of the FREQ SELECT switch.

d. The RF output power shall be attenuated a minimum of 30 dB relative to the RF output power (see 3.5.6c) with the MODE SELECT switch in the SELF TEST position for all positions of the EREQ SELECT switch.

e. The frequency stability shall be as defined in Table III.

TABLE III. STALO output readings

FREQ SELECT SWITCH POSITION	FREQUENCY STABILITY (±0.5 MHz)
9.205	9,145 MHz
9.213	9,153 MHz
9.221	9,161 MHZ
9.229	9,169 MHZ
9.237	9,177 MHz
9.245	9,185 MHz

f. The following performance requirements of MIL-S-48431 as defined in Table IV shall be met for all positions of the FREQ SELECT switch.

TABLE IV. MIL-S-48431 performance requirements

SPECIFICATION	PARAGRAPH	TITLE
MIL-S-48431	3.5.6	Spectral Purity
MIL-S-48431	3.5.6.1	Power Source Noise Spectral Density (Broadband)

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3.5.7 Self test signal.

a. All requirements shall be measured at the CAL OUT port with the MODE SELECT switch in the SELF TEST position UnleSS otherwise indicated.

b. The control assembly shall be in the CW condition unless otherwise indicated.

c. The RF output power shall be -17 dBm minimum for all positions of the FREQ SELECT switch.

d. The RF output power shall be attenuated a minimum of 80 dB relative to the RF output power (see 3.5.7c) with the MODE SELECT switch in the RADAR position for all positions of the FREQ SELECT switch.

e. The frequency stability shall be as defined in Table V.

TABLE v. Self-test readings

FREQ SELECT SWITCH POSITION	FREQUENCY STABILITY (±1.0 MHz)
9.205	9,205 MHZ
9.213	9,213 MHZ
9.221	9,221 MHZ
9.229	9,229 MHZ
9.237	9,237 MHZ
9.245	9,245 MHZ

f. All harmonic and spurious RF power shall be attenuated a minimum of 20 dB relative to the RF output power (see 3.5.7c) for all positions of the FREQ SELECT switch. Note: Measurement band width shall be the frequencies ± 180 MHz (see 3.5.7e).

g. With the control assembly in the Pulse condition, the pulse spectrum for all positions of the FREQ SELECT switch shall be as defined in Figure 1.

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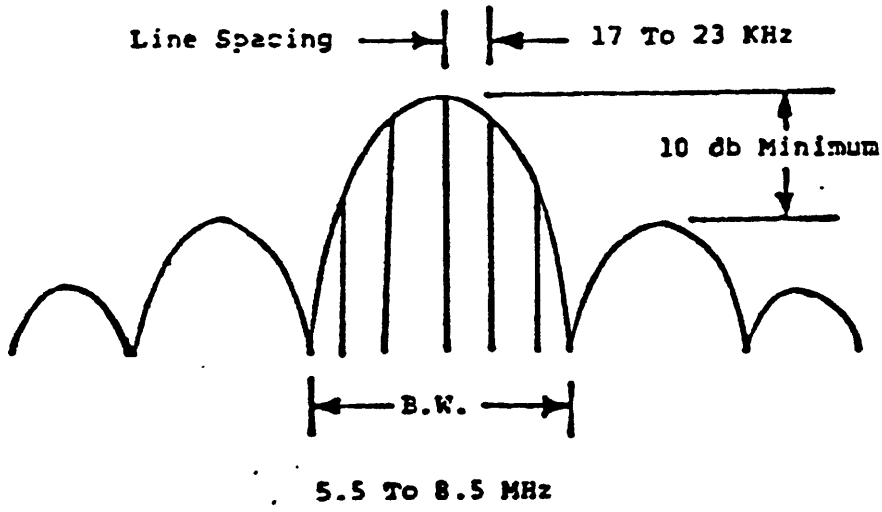


FIGURE 1. Pulse spectrum.

h. With the control assembly in the Pulse condition and the FUNCTION SELECT switch in the RF INPUT position, the RF LEVEL attenuator shall be capable of adjusting the self test signal such that the digital display (DVM) reads +1000 ± 0.0020 for all positions of theFREQ SELECT switch.

3.5.8 Cal input. The attenuation between the CAL IN port and the CAL OUT port shall be less than 6 dB over the frequency reange from 9.205 to 9.3245 MHz.

3.5.9 RF Monitor.

a. All requirements of 3.5.9 through 3.5.17p shall be accomplished with 0 dBm (CW) measured at 1A22J2 with the RF LEVEL control set to 5.

b. All requirements shall be met for all positional combinations of the FREQ SELECT switch and the RANGE SELECT switch unless otherwise indicated.

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The FUNCTION SELECT switch shall be in the RF MONITOR position.

d. The control assembly shall be in the Pulse condition unless otherwise indicated.

e. With the MODE SELECT switch in the SELF TEST position, the RF Level attenuator shall be capable of adjustment such that the digital display (DVM) reads +2.500 ± 0.050.

f. The RF LEVEL attenuator dial shall indicate 6 dB or greater at ambient conditions and 3 dB or greater at any other environmental condition specifically required by this specification.

g. With the MODE SELECT switch in the SELF TEST position, the RF LEVEL attenuator shall be adjusted such that the digital display (DVM) indicates +2.500 ± 0.020. After the initial RF LEVEL attenuator adjustment has been accomplished, the digital display (DVM) shall remain stable within a reading of +2.500 ± 0.070 for a duration of 90 ± 10 seconds.

h. The above requirements (3.5.9a through 3.5.9f) shall utilize the FREQ SELECT and RANGE SELECT combinations given in Table VI.

TABLE VI. RF monitor readings

FREQ SELECT SWITCH POSITION	RANGE SELECT SWITCH POSITION
9.205	R1
9.213	R2
9.221	R3
9.229	R4
9.237	R4
9.245	R3

3.5.10 Video output.

a. The control assembly shall be in the Pulse condition.

b. All requirements shall be measured at the VIDEO OUT port with the MODE SELECT switch in the SELF TEST position, the FUNCTION SELECT switch in the RF MONITOR position, and the RF LEVEL attenuator adjusted such that the digital display (DVM) indicates 2.500 ± 0.100 unless otherwise indicated.

c. The video output shall be a pulse having a minimum amplitude of 1.0 V (peak-to-peak) and a pulse repetition frequency between 17 KHz and 23 KHz for all positions of the FREQ SELECT switch.

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3.5.11 Data signals.

a. The following requirements shall be met with DATA INPUT cable W4 connected to control assembly connector J1. The applied voltages shall be applied to the other end of cable W4.

b. With the FUNCTION SELECT switch in the positions delineated in Table VII and voltages applied as specified in Table VII, the digital display (DVM) shall display the values specified in Table VII.

TABLE VII. Data signals

FUNCTION SELECT SWITCH POSITION	APPLIED VOLTAGE VALUE (VDC)	APPLIED (W4-PIN)	DIGITAL DISPLAY (DVM)
RANGE RATE REF (+)	+10.00 ±00.10	K (+) F (RET)	+10.00 ±00.15
RANGE REF	+20.00 ±00.10	J (+) B (RET)	+20.00 ±00.15
RANGE	+20.00 ±00.10	A (+) B (RET)	+20.00 ±00.15
RANGE RATE	+10.00 ±00.10	H (+) F (RET)	+10.00 ±00.15
RANGE RATE REF (-)	-10.00 ±00.10	G (-) F (RET)	-10.00 ±00.15

3.5.12 Signal channel.

a. The following requirements shall be met with SIGNAL INPUT cable W2 connected to control assembly connector J3.

b. With a 10 ±0.1 KHz sine wave signal with a peak-to-peak amplitude of 1.0 ±0.1 volts applied to open end of SIGNAL INPUT cable W2 (Signal-B, Ret-A), the SIGNAL CHANNEL meter shall indicate 0.5 volt min.

3.5.13 Lock on signal.

a. The following requirements shall be met with DATA INPUT cable W4 connected to control assembly connector J1.

b. With signals applied to open end of DATA INPUT cable W4 (Signal-C, Ret-D) as specified in Table VIII, the LOCK ON indicator shall indicate as specified in Table VIII.

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TABLE VIII. LOCK ON INDICATOR FUNCTIONS

APPLIED SIGNAL	LOCK ON INDICATOR
+5.0 ± 0.1 vdc	Not Lighted
+5.0 ± 0.1 vdc	Lighted (Indicator Depressed)
Open	Lighted
+1.0 ± 0.1 Vdc	Lighted

3.5.14 Elapsed time indicator. With POWER SUPPLY cable W1 connected to control assembly connector J2 and +28 ± 0.6 Vdc applied to cable W1, W1-C (+), W1-D (ret), the elapsed time indicator shall have an audio sound indicating activation and shall display operational time within ±33% in a three hour continuous-period.

3.5.15 Range and Phase insertion.

a. The control assembly shall be in the Pulse condition.

b. With the FUNCTION SELECT switch in the RF MONITOR position, the RANGE SELECT switch in the R3 position, and the PHASE CONTROL at midrange, the RF LEVEL attenuator shall be adjusted such that the digital display (DVM) indicates +2.500 ± 0.020.

c. With the control assembly in the Pulse condition, the digital display (DVM) shall not vary more than ± 1.200 for RANGE SELECT switch positions R1, R2, and R4.

d. With the control assembly in the Pulse condition, the digital display (DVM) shall not vary more than ±0.500 when the PHASE CONTROL is varied from one extreme (counter clockwise) position to the other extreme (clockwise) position.

e. All requirements shall be measured for all positions of the FREQ SELECT switch.

3.5.16 Clutter signal characteristics.

a. All clutter signal characteristics are to be measured with an internal RF power calibration (Pulse) accomplished prior to (within 30 seconds) of any measurement unless otherwise indicated.

b. An internal RF power calibration (Pulse) shall be accomplished by adjusting the RF LEVEL dial such that the digital display (DVM) indicates +2.500 ± 0.025 with the RF FUNCTION switch in the RF MONITOR position.

c. All clutter signal characteristics shall be measured with the MODE SELECT switch in the SELF TEST position unless otherwise indicated.

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d. Unless otherwise indicated, all clutter signal characteristics shall be measured for all positions of the FREQ SELECT switch with the control asssembly in the CW condition.

e. All clutter signal characteristics shall be measured at one end of cable W5 with the other end of cable W5 connected to the RADAR port of the control assembly unless otherwise indicated.

f. With the CLUTTER switch in the OFF position, CLUTTER LEVEL dial set to 5 (five), and the CLUTTER BALANCE attenuator set to minimum, the clutter signal (CW) shall be less than -90 dBm.

g. With the CLUTTER switch in the ON position and the CLUTTER LEVEL dial set to 20, the CLUTTER BALANCE attenuator shall be capable of adjusting the clutter signal (CW) from -18 dBm to -22 dBm.

h. All harmonically related clutter power shall be attenuated a minimum of 40dB relative to the clutter signal (CW), over a frequency range of clutter signal ±180 MHz.

i. All spurious related clutter power shall be attenuated a minimum of 60 dB relative to the clutter signal (CW) over a frequency reange of clutter signal ±180 MHz.

j. For FREQ SELECT switch positions 9.205 and 9.245 only, the clutter signal (CW) shall be attenuated referenced to the CLUTTER LEVEL dial over the dial range from 5 to 65 within the accuracy requirements given in Table IX. An internal RF power calibration (pulse) is not required.

TABLE IX. Clutter signal attenuation

DIAL READING IN dB	ROOM TEMP TOLERANCE IN dB
15	±1.5
25	±1.5
35	±1.5
45	±1.5
55	±1.5
65	±2.0

k. The clutter signal (CW) shall be 60 +1.0 MHz higher in frequency than the STALO output frequency defined in 3.5.6e.

l. With the CLUTTER LEVEL dial at 20, cable W5 connected to the SELF TEST port of the control assemlby, and CLUTTER BALANCE attenuator adjusted such that with the FUNCTION SELECT switch in the RF SELF TEST position the digital display (DVM) indicates the RF SELF TEST calibration value (Pulse), the clutter signal (CW) shall be -20.0 ±0.1 dBm for FREQ SELECT switch positions 9.229, 9.205, and 9.245 only.

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m. With the CLUTTER LEVEL dial set at 5, an increase of 5 (higher dial setting) in the RF LEVEL dial from its internal RF power calibration setting shall cause a 5 ± 1.0 dB reduction in the clutter signal (CW).

n. The clutter signal (Pulse) shall have a pulse spectrum as delineated in Figure 1 with the RF LEVEL dial set at its internal RF power calibration (Pulse) reference setting and shall maintain the pulse spectrum characteristics when the RF LEVEL dial is increased (higher dial setting) an increment of 5.

3.5.17 Target signal characteristics.

a. All target signal characteristics are to be measured with an internal RF power calibration (Pulse) accomplished prior to (within 30 seconds) of any measurement unless otherwise indicated.

b. An internal RF power calibration (Pulse) shall be accomplished by adjusting the RF LEVEL attenuator such that the digital display (DVM) indicates +2.500 ± 0.025 with the RF FUNCTION switch in the RF MONITOR position.

c. All target signal characteristics shall be measured with the MODE SELECT switch in the SELF TEST position unless otherwise indicated.

d. All target signal characteristics shall be measured for all positions of the FREQ SELECT switch with the control assembly in the CW condition unless otherwise indicated.

e. All target signal characteristics shall be measured at one end of cable W5 with the other end of cable W5 connected to the RADAR port of the control assembly unless otherwise indicated.

f. With the TARGET switch in the OFF position, TARGET LEVEL attenuator dial set to 27, and the TARGET BALANCE attenuator set to minimum, the target signal (CW) shall be less than -90 dBm.

g. With the TARGET switch in the ON position and the TARGET LEVEL attenuator dial set to 27, the TARGET BALANCE attenuator shall be capable of adjusting the target signal (CW) from -22 dBm to -28 dBm.

h. The target (main spectral line) signal (CW) shall be 3.5 KHz to 5.5 KHz higher in frequency than the clutter signal as defined in 3.5.16k.

i. The target spectrum (CW) within a 200 KHz bandwidth shall have all spurious and harmonically related sidebands suppressed a minimum of 20 dB from the target signal (main spectral line).

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j. For FREQ SELECT switch positions 9.205 and 9.245 only, the target signal (CW) shall be attenuated referenced to the TARGET LEVEL attenuator dial over the dial range from 27 to 87 within the accuracy requirements given in Table X. An internal RF power claibration (pulse) is not required.

TABLE X. Target signal attenuation

DIAL READING IN dB	ROOM TEMP TOLERANCE IN dB
37	±1.5
47	±1.5
57	±1.5
67	±1.5
77	±1.5
87	±1.5

k. For FREQ SELECT switch in the 9.229 position only, with the TARGET switch and the TARGET FREEZE switch in the positions indicated, the target signal when envelope detected shall be as indicated in Table XI.

TABLE XI. Target signal outputs.

TARGET SWITCH	TARGET FREEZE SWITCH	TARGET SIGNAL OUTPUT
OFF	OFF	None (less than 10%)
OFF	ON	None (less than 10%)
ON	OFF	Pulse (100% amplitude ref.)
ON	ON	Figure 2

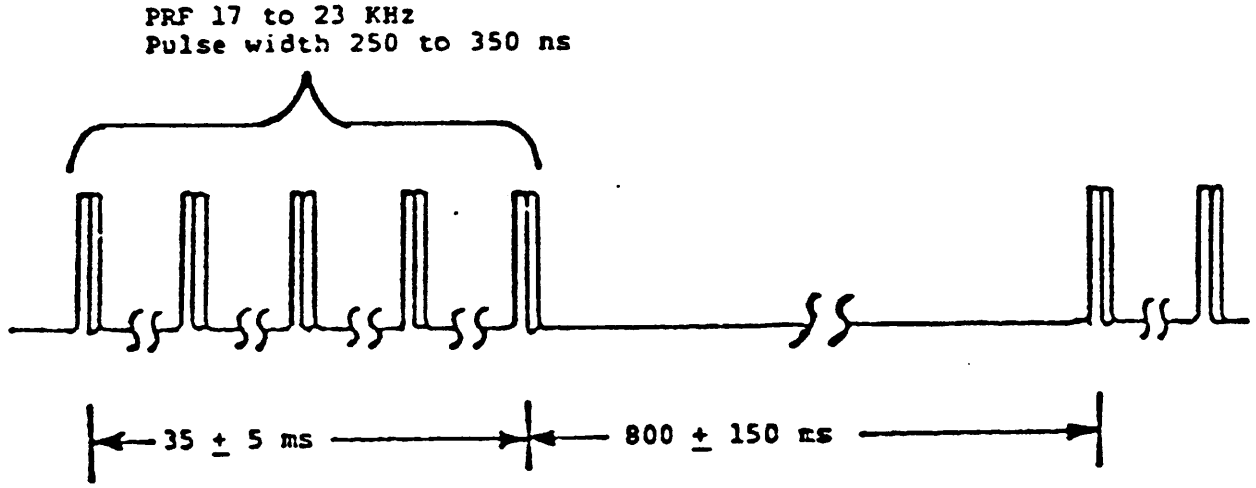


FIGURE 2. Target signal output.

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l. With the CLUTTER LEVEL attenuator dial at 25 (after the requirements of 3.5.16 step 1 have been met), the TARGET LEVEL attenuator dial at 27, cable W5 connected to the SELF TEST port of the control assembly, and the FUNCTION SELECT switch in the RF SELF TEST position, the TARGET BALANCE attenuator shall be capable of adjustment such that the digital display (DVM) indicates the same value within ±0.010 for the target signal (Pulse) as the clutter signal (Pulse).

m. After the requirements of step 1., above, have been met the difference between the target signal (CW) and clutter signal (CW) shall be 1.5 dB max with CLUTTER LEVEL dial set to 27.

n. After the requirements of step 1., above, have been met the clutter signal shall vary less than ±1.0 dB with the TARGET switch ON or OFF.

o. With the TARGET LEVEL attenuator dial set at 27, an increase of 5 (higher dial setting) in the RF LEVEL attenuator dial setting from its internal RF power calibration setting shall cause a 5 ±1.0 dB reduction in the target signal (CW).

p. The target signal (Pulse) shall have a pulse spectrum as delineated in Figure 1 with the RF LEVEL dial set at its internal RF power calibration (Pulse) reference setting and shall maintain the pulse spectrum characteristics when the RF LEVEL dial is increased an increment of 5 (higher dial setting).

3.5.18 Calibration data.

3.5.18.1 RF input.

a. RF INPUT calibration data shall be determined such that when an RF input peak power between 0.125 KW and 2.0 KW is applied to the end of cable W5 with the other end of cable W5 connected to the control assembly RADAR port, the FUNCTION SELECT switch in the RF INPUT position and the RF LEVEL attenuator adjusted such that the digital display (DVM) indicates +2.500 ±0.020; that the RF INPUT calibration number when added to the RF LEVEL attenuator dial setting measures the RF input peak power within ±0.5 dB for RF input frequencies between 9,205 and 9,245 MHz with a PRF of 20.0 ±0.2 KHz and a pulse width of 280 ±20 ns.

b. The RF INPUT calibration value shall be within the limits of 30.0 and 52.0.

c. The RF INPUT calibration data is to be single value in dB (xx.x) applicable for all conditions and shall be recorded on the calibration data card located in the control assembly panel flyleaf.

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3.5.18.2 RF self test.

a. RF SELF TEST calibration data shall be determined such that with a -20.0 ± 0.1 dBm peak power signal injected directly into the control assembly SELF TEST port, the digital display (DVM) shall indicate the RF SELF TEST calibration value within ± 0.100 with the FUNCTION select switch in the RF SELF TEST position. The injected RF peak power shall have the following characteristics: frequency 9.229 MHZ, PRF 17 to 23 KHz, and pulse width 300 ± 50 ns.

b. The RF SELF TEST calibration value shall be within the limits of 1.50 and 2.50.

c. The RF SELF TEST calibration data is to be a single value digital display (DVM) indication (x.xx) applicable for all conditions and shall be recorded on the calibration data card located in the control assembly panel flyleaf.

3.5.18.3 Range.

a. RANGE calibration data shall be determined for RANGE SELECT switch positions R1, R2, R3 and R4. The range delay (pulse delay) shall be determined in nanoseconds for an RF pulse starting at the CAL OUT port and stopping at the open end of cable W5 (other end of cable W5 Connected to RADAR port). The range delay shall be measured with the CLUTTER switch in the ON position and the TARGET switch in the OFF position. The RF input (start) pulse shall have the following characteristics: frequency 9.229 MHz, PRF 17 KHz to 23 KHz, pulse width 250 ns to 350 ns.

b. The range delay shall vary no more than ±25 nanoseconds from the RANGE calibration data in nanoseconds for all conditions.

c. The range delay for R1, R2, R3, and R4 shall be converted from nanoseconds to meters using 0.15 meter equals 1.0 nanoseconds.

d. The RANGE calibration data shall be within the following limits: R1:300 ± 20, R2:400 ± 20, R3:1575 ± 20 and R4:4715 ± 20.

e. The range delay in meters (xxxx.) shall be recorded as the RANGE calibration data on the calibration data card located in the control assembly panel flyleaf.

3.5.19 Radar Test.

a. The control assembly shall be capable of performing a complete RADAR TEST by measuring the following parameters of an AN/VPS-2 Vulcan Radar as given in Table XII for all positions of the FREQ SELECT switch unless otherwise indicated.

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TABLE XII. AN/VPS-2 parameters.

AN/VPS-2 PARAMETER	ROOM TEMP ACCURACY	ACCURACY OVER TEMP RANGE
RF Peak Transmitted Power	±1.0 dBm	±1.5 dBm
Minimum Discernible Signal (MDS)	±1.0 dBm	±1.5 dBm
Sub Clutter Visibility (SCV)	±1.5 dB	±2.0 dB
AGC Threshold	±1.0 dB	ld.5 dB
AGC Linearity	±1.5 dB	±2.0 dB
Range Accuracy	±7 Meters	±12 Meters
Range Rate Ref (+/-)	±0.050 Vdc	±0.075 Vdc

b. During environmental service conditions, temperature, operating per 3.3:3.1 the FREQ SELECT switch shall be in the 9.245 position.

c. The AN/VPS-2 Vulcan Radar shall be maintained in a calibration status by external independent verification of parameter data, by periodic verification utilizing the factory test rack, or by verification of calibration status utilizing an AN/VPM-2 system having met all requirements of MIL-P-48606 and this specification.

3.6 Environmental Stress Screening (ESS). ESS shall be performed on each new control assembly produced and shall consist of two non-operating screening procedures. The control assembly shall meet the requirements of 3.4 and 3.5 at standard ambient conditions after each procedure.

3.6.1 Screening procedures.

3.6.1.1 Temperature cycling. The control assembly shall be exposed to three (3) ambient temperature cycles between -60 degrees F and +160 degrees F with a minimum change rate of 5.5 degrees F per minute and a minimum dwell time of 60 minutes at each extreme in the cycle. The first cycle shall start at standard ambient conditions and the first extreme shall be +160 degrees F. It shall take 3.33 hours to complete 1 cycle, 10 hours total. If the minimum change rate is exceeded, the dwell times shall be equally increased accordingly.

3.6.1.2 Vibration. The control assembly shall be exposed to vibrations using the same conditions as 3.3.1 except that the cycling time shall be reduced to 20 minutes per axis.

3.6.2 Failures after a screening procedure. Each failure shall be recorded, categorized, and corrected as follows:

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3.6.2.1 Workmanship failures. Workmanship failures shall be corrected without repeating the screening procedure that caused the failure.

3.6.2.2 Component/subassembly failures. Component/subassembly failures shall be corrected by replacing the failed part with a new part that has been screened up to-the procedure at which the failure occurred. The screening procedure that caused the part failure shall then be repeated on the control assembly.

3.7 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein

3.8 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (task) F of paragraph 1.2 of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor’s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 deg to +90 deg F.
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg.

4.3 First article inspection.

4.3.1 First article submission. The requirement for first article and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract and the applicable drawing. The sample for the first article tests shall consist of one (1) control assembly manufactured in the same reamer, using the same materials, equipment, processes and procedures in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.3.1.1 Government testing. When the Government is responsible for conducting first article tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and the applicable drawings. The contractor shall submit a record of this inspection with the sample including certified Test Report (CTR) or Certificate of Conformance (COC) for the materials.

4.3.1.2 Contractor testing. When the Contractor is responsible for conducting first article tests, the sample shall be inspected by the contractor for all the requirements of the contract, all of the examinations and tests specified in Table XIII (Functional Test) and Table XIV (Sepcal Sampling) herein and to all requirements of the applicalble drawings. The sample and a record of this inspection, including CTR/COC, shall be submitted for the Governement approval. The Government reserves the right to witness the Contractor's inspection.

4.3.2 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term “inspection lot” is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to section 3 herein shall be performed in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with sections herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table XIII shall be performed on a 100% basis.

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TABLE XIII. Functional tests
(100% Inspection)

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
Critical:		**None**	**N/A**
Major:			
101.	**Electrical**	**3.4**	**4.7.2**
102.	**Performance**	**3.5**	**4.7.3**
103.	**ESS**	**3.6**	**4.7.4**
Minor:		**None**	**N/A**

4.5 Special sampling. One control assembly out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table XIII and shall then meet the tests in Table XIV. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE XIV. Special sampling

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
Critical:		None	N/A
Major:			
104.	Vibration	3.3.1	4.7.1.1
105.	Altitude	3.3.2	4.7.1.2
106.	Temperature	3.3.3	4.7.1.3
107.	Humidity	3.3.4	4.7.1.4
108.	Immersion	3.3.5	4.7.1.5
Minor:		None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production have been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where are contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

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4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval In accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table XV shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE XV. Test equipment list

ITEM	NUMBER	DESCRIPTION
1.	Power Supply Unit 2	P/N 11748133
2.	Power Supply	0 to 30 Volts, 15 Amps min. HP Model #6434B or equivalent.
3.	Cables	P/N 11748213 P/N 11748214 P/N 11748215 P/N 11748216 P/N 11748144
4.	Power Supply	0 to 40 volts, 0.5 Amps
5.	Function Generator	Systron Donner, Model #415 or equivalent.

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TABLE XV. Test equipment list (continued)

ITEM NUMBER		DESCRIPTION
6.	Frequency Counter	HP 5345A with X-Band Plug-In or equivalent.
7.	RF Amplifiers (2)	Part No. 11748155.
8.	X-Band Rotary Vane Waveguide Variable Attenuator	Waveline Model #60562 or equivalent.
9.	Two Coaxial Detectors	Aertech Model #8596BZS or equivalent.
10.	Waveguide to Coaxial Adapter	Omin Spectra Model #2000-6254-00 or equivalent.
11.	Precision Attenuator Set	Weinschel Engineering Model #9168 or equivalent.
12.	Spectrum Analyzer	Tektronix, Model #491 or equivalent.
13.	Power Meter	HP 432A or equivalent.
14.	Oscilloscope	Tektronix, Model #454 or equivalent.
15.	Signal Generator, X-Band	Alfred Model #650 with Model #655K plug-in or equivalent.
16.	Digital Multimeter	5 1/5 digits minimum, true-RMS AC Fluke Model #8842A/09 or equivalent.
17.	Radar Set AN/VPS-2	P/N 10549058
18.	Noise Tester	RDL Model #TE02X11 or equivalent.

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The control assembly shall be attached to the vibration generator directly or with a fixture and securely held

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by its normal means of attachment. After vibration, the control assembly shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.2.

4.7.1.3 Temperature.

4.7.1.3.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.3. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.3.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II; Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.3. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Immersion (non-operating). This test shall be performed as specified in Procedure 1, Method 512 of MIL-STD-810 as modified by 3.5.5 herein. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.7.3 Performance.

4.7.3.1 Initial conditions.

a. Connect cable W1 at both ends and comect cable W3 to the control assembly only as shown in figure 3. If individual power supplies are utilized (see 3.5.1) connect individual power supplies in lieu of Power Supply Unit 2 in figure 3.

b. Set control assembly front panel controls as specified in Table XVI.

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TABLE XVI. Control settings.

POWER	ON
FUNCTION SELECT	VOLTAGE
VOLTAGE SELECT	+24
RF LEVEL	20
FREQ SELECT	9.205
MODE SELECT	SELF TEST
PHASE CONTROL	MID RANGE
CLUTTER	OFF
CLUTTER LEVEL	20
RANGE SELECT	R1
TARGET LEVEL	27
TARGET FREEZE	OFF
TARGET	OFF

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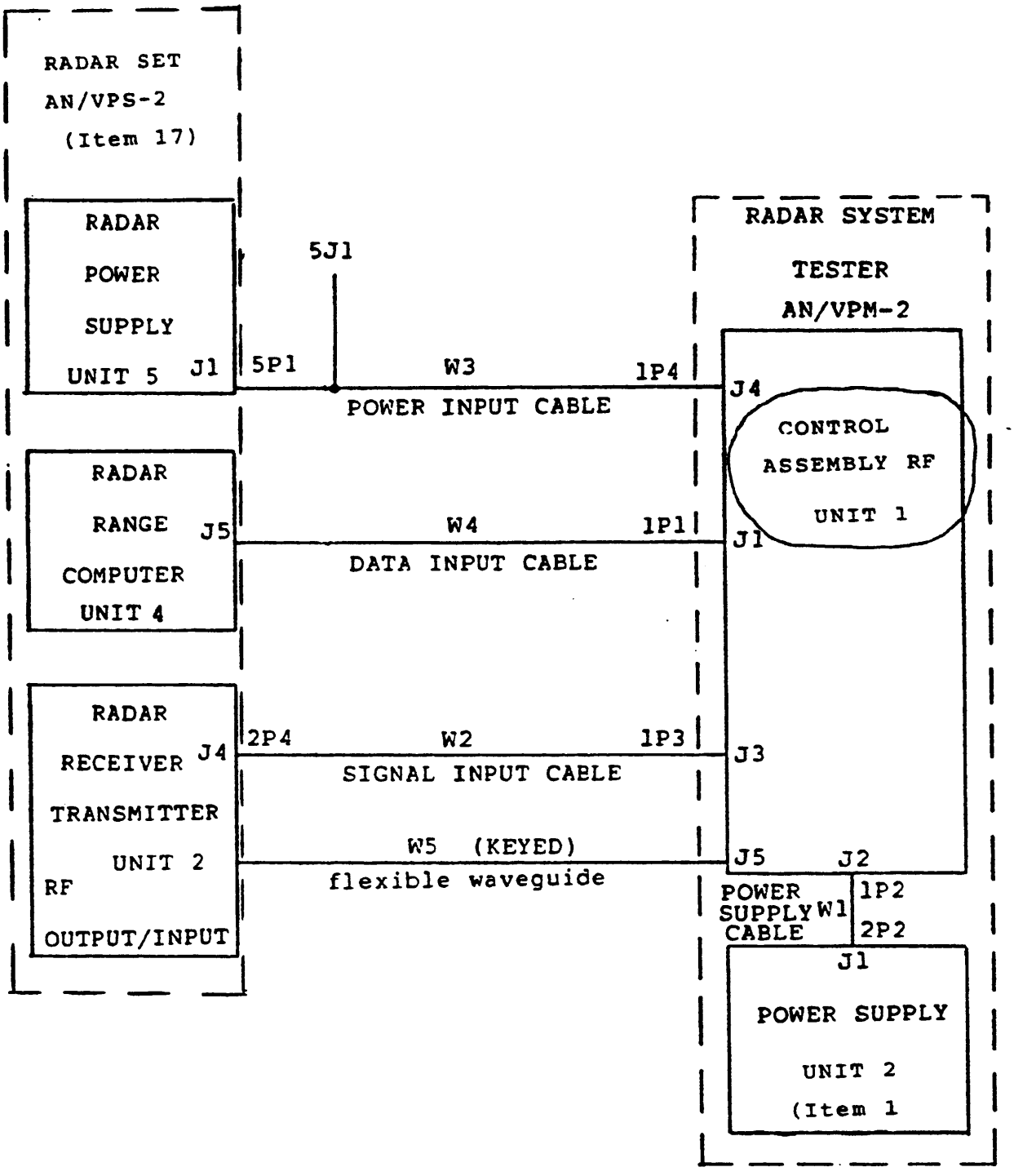


FIGURE 3. Radar system tester setup.

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c. place the control assembly into the CW condition (see 3.5.3.1) unless otherwise indicated.

d. Using power supply (item 2, Table XV) apply +24 ± 0.2 Vdc(unless otherwise specified) to power input cable W3 (either 5J1 or 5P1) pins A (+), B(ret), C (Cnd).

e. Turn POWER switch to ON position.

f. Allow an initial warm-up as defined in 3.5.2.

4.7.3.2 Voltage monitor.

a. Perform initial conditions per 4.7.3.1.

b. Place the FUNCTION SELECT switch in the VOLTAGE position.

c. Using the DMM (item 16, Table XV) measure and record the applied voltage as specified in Table I.

d. Measure and record the digital display (DVM) indication for each position of the VOLTAGE SELECT switch per Table 1.

e. The difference between the applied voltage and the digital Display (DVM) indication is the tolerance, and shall be as specified in 3.5.4a.

f. Place the VOLTAGE SELECT switch in the +24 position.

g. Measure and record the voltage between W1-H (+) and W1-I (ret) per Table I.

4.7.3.3 Voltage input monitor.

a. Perform initial conditions per 4.7.3.1.

b. Place the FUNCTION SELECT switch in the VOLTAGE position.

c. Place the VOLTAGE SELECT switch in the VOLT TP position.

d. Apply inputs as specified in Table II to the INPUT jack and GNC jack (see 3.5.5).

e. Measure and record the applied inputs using the DMM (item 16, Table XV) and digital display (DVM) indications per Table II.

4.7.3.4 STALO output.

a. Perform initial conditions per 4.7.3.1

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b. Place the control assembly in the Pulse condition see 3.5.3.2).

c. Remove the 50-ohm termination from the STALO OUT port and in its place connect a 10 dB coaxial attenuator and RF power meter. Set MODE SELECT switch to RADAR position.

d. Measure and record the RF output power for all positions of the FREQ SELECT switch per 3.5.6c.

e. Remove the RF power meter and connect a 30 dB coaxial attenuator and a spectrum analyzer.

f. Measure and record the attenuation with the MODE SELECT switch in the SELF TEST position relative to the attenuation with the MODE SELECT switch in the RADAR position for all positions of the FREQ SELECT switch per 3.5.6d.

g. Remove the spectrum analyzer and the 30 dB coaxial attenuator and connect a frequency counter and a 10 dB coaxial attenuator. Place the MODE SELECT switch in the RADAR position.

h. Measure and record the STALO output frequency for all positions of the FREQ SELECT switch per 3.5.6e and Table 111.

i. Remove the frequency counter and 10 dB coaxial attenuator and connect equipment as specified in MIL-S-48431 or Noise Tester (item 18, Table XV).

j. Measure and record the spectral purity and power source spectral density as specified in 3.5.6f and Table IV.

k. Remove test equipment from STALO OUT port and reconnect 50-ohm termination.

4.7.3.5 Self test signal.

a. Perform initial conditions per 4.7.3.1.

b. Place the control assembly in the CW condition.

c. Remove the 50-ohm termination from the CAL OUT port and in its place connect an RF power meter with appropriate adapters.

d. Measure and record the RF output power for all positions of the FREQ SELECT switch per 3.5.7c.

e. Remove the RF power meter and connect a spectrum analyzer.

f. Measure and record the attenuation with the MODE SELECT switch in the SELF TEST position relative to the RF output power measured in 4.7.3.5.d with the MODE SELECT switch in the RADAR position for all positions of the FREQ SELECT switch per 3.5.7d.

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g. Place the MODE SELECT switch in the SELF TEST position.

h. Measure and record the attenuation of all harmonic and spurious RF power for all positions of the FREQ SELECT switch per 3.5.7f.

i. Place the control assembly in the Pulse condition.

j. The pulse spectrum for all positions of the FREQ SELECT switch shall be as defined in Figure 1 per 3.5.7g.

k. Remove the spectrum analyzer and connect a frequency counter in its place.

l. Place the control assembly in the CW condition.

m. Measure and record the self test output frequency for all positions of the FREQ SELECT switch per 3.5.7e and Table V.

n. Remove the frequency counter and replace the 50-ohm termination on the CAL OUT port.

o. Place the control assembly in the Pulse condition.

p. Place the FUNCTION SELECT switch in the RF INPUT position.

q. Adjust the RF LEVEL attenuator such that the digital display indicates +1.000 ± 0.020 for all positions of the FREQ SELECT switch per 3.5.7.h.

4.7.3.6 CAL input.

a. Perform initial conditions per 4.7.3.1.

b. Remove the 50-ohm termination from the CAL IN port and the CAL OUT port.

c. Inject a low level (approximately -5 dBm) signal into the CAL IN port at a frequency of 9,205 MHz ± 1 MHz.

d. Measure and record the insertion loss at the CAL OUT port per 3.5.8.

e. Repeat steps c and d above with the low level signal at a frequency of 9,245 MHz ± 1MHz.

f. Replace the 50-ohm terminations.

4.7.3.7 RF monitor.

a. Perform initial conditions per paragraph 4.7.3.1 and adjust RF LEVEL control to 5.

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b. Connect power meter to 1A22J2.

c. Place control assembly in the CW condition and adjust 1A22 for 0 dBm reading on the power meter.

d. Place the FUNCTION SELECT switch in the RF MONITOR position.

e. Place the control assembly in the Pulse condition.

f. Place the RANGE SELECT switch in the R1 position.

g. Place the FREQ SELECT switch in the 9.205 position.

h. Adjust the RF LEVEL attenuator until the digital display (DVM) indicates +2.500 ±0.050 per 3.5.9f. Record RF LEVEL measurement.

i. Repeat step h for each of the remaining switch combinations listed in Table VI.

j. Adjust the RF LEVEL attenuator until the digital display (DVM) indicates +2.500 ± 0.020.

k. Without further adjustment, the digital display (DVM) shall indicate as specified in 3.5.9g.

4.7.3.8 Video output.

a. Perform initial conditions per 4.7.3.1.

b. Place the control assembly in the Pulse condition.

c. Place the FUNCTION SELECT switch in the RF MONITOR position.

d. Remove the 50-ohm termination from the VIDEO OUT port and connect an oscilloscope in its place.

e. Adjust the RF LEVEL attenuator per 3.5.10b.

f. Measure and record the video output per 3.5.10c for all positions of the FREQ SELECT switch.

g. Remove the oscilloscope and reconnect the 50-ohm termination to the VIDEO OUT port.

4.7.3.9 Data signals.

a. Perform the initial conditions per 4.7.3.1.

b. Connect DATA INPUT cable W4 to connector J1 on the control assembly. Connect power supply to other end of W4.

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c. With the FUNCTION SELECT switch in the positions specified in Table VII, and applied voltages specified in Table VII, measure and record the digital display (DVM) per 3.5.11b.

4.7.3.10 Signal channel.

a. Perform the initial conditions per 4.7.3.1.

b. Connect SIGNAL INPUT cable W2 to control assembly connector J3. Connect signal generator to other end of cable W2.

c. Apply signals per 3.5.12b.

d. Measure and record the signal charnel meter reading per 3.5.12b.

4.7.3.11 Lock on signal.

a. Perform the initial conditions per 4.7.3.1.

b. Connect data input cable W4 to control assembly connector J1. Connect power supply to other end of cable w4.

c. Apply signals specified in Table VIII per 3.5.13b.

d. Record the LOCK ON indicator condition per Table 3.5.13b.

4.7.3.12 Elapsed time indicator. This test can be performed during any three consecutive hours of ruining time. Record indication on elapsed time indicator at the start and conclusion of three consecutive hours of operation per 3.5.14.

4.7.3.13 Range and phase insertion.

a. Perform the initial conditions per 4.7.3.1.

b. Place the control assembly in the Pulse condition.

c. Place the FUNCTION SELECT switch in the RF MONITOR position.

d. Place the RANGE SELECT switch in the R3 position.

e. Place the PHASE CONTROL knob in the approximate mid-range position.

f. Adjust the RF LEVEL attenuator such that the digital display (DVM) indicates +2.500 ± 0.020.

g. Measure and record the digital display (DVM) indications for RANGE SELECT positions R1, R2, R3, and R4 without further adjustment of the RF LEVEL attenuator.

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h. Vary the PHASE CONTROL knob from one extreme (counterclockwise) position to the other extreme (clockwise) position.

i. Record the digital display (DVM) indications at each extreme position per 3.5.15d.

j. Repeat steps d through i above for all positions of the FREQ SELECT switch.

4.7.3.14 Clutter signal characteristics.

a. Perform the initial conditions per 4.7.3.1.

b. Perform RF power calibration per 3.5.16a and 3.5.16b as required.

c. Connect a spectrum analyzer and waveguide variable attenuator to the open end of cable W5.

d. Set the CLUTTER LEVEL dial to 5.

e. Set the CLUTTER BALANCE attenuator to a minimum (completely counter clockwise).

f. Measure and record the clutter signal per 3.5.16f for all positions of the FREQ SELECT switch.

g. Set the CLUTTER LEVEL dial to 20.

h. Place the CLUTTER switch to ON position.

i. Adjust the CLUTTER BALANCE attenuator per 3.5.16g.

j. Measure and record the clutter signal per 3.5.16g for all positions of the FREQ SELECT switch.

k. Measure and record the harmonically related clutter power per 3.5.16h for all positions of the FREQ SELECT switch.

l. Measure and record the spurious related clutter power per 3.5.16i for all positions of the FREQ SELECT switch.

m. Set the RF LEVEL dial to 15.

n. Set the FREQ SELECT switch to the 9.205 position.

o. Set the CLUTTER LEVEL dial to 5.

p. Measure and record the clutter signal level versus the CLUTTER LEVEL dial reading in dial increments of 10 per 3.5.16j and Table IX.

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q. Repeat steps o and p with the FREQ SELECT switch in the 9.245 position.

r. Remove the spectrum analyzer and waveguide variable attenuator from the open end of cable W5 and connect a frequency counter.

s. Measure and record the clutter signal frequency for all positions of the FREQ SELECT switch per 3.5.16k.

t. Connect an RF power meter to cable w5.

u. Place the FREQ SELECT switch in the 9.229 position.

v. Set the CLUTTER LEVEL dial at 20 and FUNCTION SELECT switch to RF MONITOR. Adjust RF LEVEL control for a reading of 2.500 ± 0.020 on the digital display (DVM).

w. Adjust the CLUTTER BALANCE attenuator such that the clutter signal (CW) is -20.0 ± 0.1 dBm.

x. Remove the RF power meter and connect cable W5 to the control assembly SELF TEST port.

y. Place the FUNCTION SELECT switch in the SELF TEST position.

z. Place the control assembly in the Pulse condition.

aa. measure and record the digital display (DVM) per 3.5.16j.

ab. Repeat steps t through aa, above, for FREQ SELECT positions 9.250-and 9.245.

ac. Connect an RF power meter to cable W5.

ad. Set the CLUTTER LEVEL dial at 5.

ae. Measure and record the change in clutter signal (CW) when the RF LEVEL dial is increased by an increment of 5, from its internal RF power calibration reference dial setting per 3.5.16m for all positions of the FREQ SELECT switch.

af. Remove the RF power meter and connect a 20 dB coaxial attenuator and spectrum analyzer to cable W5.

ag. Place the control assembly in the Pulse condition.

ah. Measure and record the pulse spectrum per 3.5.16n for all positions of the FREQ SELECT switch.

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4.7.3.14 Target signal characteristics.

a. Perform the initial conditions per 4.7.3.1.

b. Perform RF power calibration per steps 3.5.17a and 3.5.17b as required.

c. Connect a spectrum analyzer and waveguide variable attenuator to the open end of cable W5.

d. Set the TARGET LEVEL dial to 27.

e. Set the TARGET BALANCE attenuator to a minimum (completely counterclockwise).

f. Measure and record the target signal per 3.5.17f for all positions of the FREQ SELECT switch.

g. Place the TARGET switch to ON position.

h. Adjust the TARGET BALANCE attenuator per 3.5.17g.

i. Measure and record the target signal per 3.5.17g for all positions of the FREQ SELECT switch.

j. Place the CLUTTER switch to the ON position and adjust the clutter signal (within 10 dB) of a target signal reference.

k. Measure and record the frequency difference between the clutter signal and the target signal per 3.5.17h for all positions of the FREQ SELECT switch.

l. Place the CLUTTER switch to the OFF position and TARGET LEVEL to 27.

m. Measure and record the harmonically and spurious related sidebands with the control assembly in CW condition for the requirements of 3.5.17i for all positions of the FREQ SELECT switch.

n. Set the RF LEVEL dial to 15.

o. Set the FREQ SELECT switch to the 9.205 position.

p. Set the TARGET LEVEL dial to 27.

q. Measure and record the target signal level versus the TARGET LEVEL dial reading in dial increments of 10 per 3.5.18j and Table X.

r. Repeat steps p and q, above, with the FREQ SELECT switch in the 9.245 position.

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s. Remove the spectrum analyzer and connect a diode detetor RF amplifier, and oscilloscope.

t. With the FREQ SELECT switch in the 9.299 position measure and record the target signal output with the TARGET FREEZE and TARGET switch ocmbinations per 3.5.18k and Table XI.

u. Remove the RF amplifier and diode detector and oscilloscope from cable W5.

v. Connect cable W5 to the control assembly SELF TEST port.

w. Place the CLUTTER switch in the ON position, the TARGET switch in the OFF position.

x. Place the control assembly in the Pulse condition.

y. Adjust the CLUTTER BALANCE attenuator until the digital display (DVM) indicates the value per 3.5.16 step 1. (a) (RF self test calibration value).

z. Place the CLUTTER LEVEL dial at 25 and the TARGET LEVEL dial at 27 with CLUTTER switch ON and TARGET switch OFF.

aa. Record the digital display (DVM) indication.

ab. Place the CLUTTER switch in the OFF position and the TARGET switch in the ON position.

ac. Adjust the TARGET BALANCE attenuator until the digital display (DVM) indicates the value recorded in step z. above. The digital display shall indicate per 3.5.17 step 1.

ad. Disconnect cable W5 from the SELF TEST port and connect a spectrum analyzer to the open end of cable W5.

ae. Place the control assembly in the CW condition. Set the CLUTTER LEVEL dial at 27. Turn the TARGET and CLUTTER switches to ON. Measure and record the difference between the TARGET and CLUTTER signal levels per 3.5.17m.

af. Turn the TARGET switch to OFF. Measure and record the change in CLUTTER signal per 3.5.17n.

ag. Set the TARGET LEVEL dial at 27 and the TARGET switch to ON.

ah. Measure and record the change in target signal (CW) when the RFL LEVEL dial is increased by an increment of 5 from its internal RF power calibration reference dial setting per 3.5.17o.

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ai. Place the control assembly in the Pulse condition.

aj. Measure and record the pulse spectrum per 3.5.17p for all positions of the FREQ SELECT switch.

4.7.3.16 Calibration data.

a. Perform the initial conditions per 4.7.3.1.

b. Place the FUNCTION SELECT switch in the RF INPUT position.

c. Connect cable W5 to the RADAR port.

d. Inject an RF peak power as specified in 3.5.18.1a having a frequency of 9,229 ± 1 MHz into the open end of cable W5.

e. Adjust the RF LEVEL attenuator until the digital display (DVM) indicates +2.500 ± 0.020.

f. Determine the RF INPUT calibration number by subtracting the RF LEVEL attenuator dial setting from the RF peak power input. The value shall be within the requirements of 3.5.18.1b.

g. Record RF INPUT calibration data per 3.5.18.1c.

h. Place the FUNCTION SELECT switch in the RF SELF TEST position.

i. Inject into the control assembly SELF TEST port an RF peak power having characteristics as specified in 3.5.18.2a at a frequency of 9,229 ± 1 MHz.

j. The digital display (DVM) shall indicate a value between +1.000 and +2.500 per 3.5.18.2b.

k. Record the value (step j, above), as the RF self test calibration data per 3.5.19.2c.

l. Place the MODE SELECT switch in the SELF TEST position and the FREQ SELECT switch in the 9.229 position.

m. place the control assembly in the Pulse condition.

n. Place the RANGE SELECT switch in the R1 position.

o. Connect a detector and an RF amplifier to the CAL OUT port and connect video end of detector through a coaxial T to CH A START on counter. Connect the other arm of the coaxial T to CH A of oscilloscope. Ensure oscilloscope is DC COUPLED.

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p. Connect another detector and RF amplifier to open end of cable WS. (use W/G to Coax Adapter.) Connect video end of detector through a coaxial T to CH B STOP on counter. Connect the other arm of the coaxial T to CX B of oscilloscope. Ensure oscilloscope is DC COUPLED.

q. Set CLUTTER switch to ON position and TARGET switch to OFF position.

r. Adjust oscilloscope until pulse display is positioned with the 50% point of the pulse amplitude on the center graticule of the vertical display.

s. Disconnect detector lead from oscilloscope CH A.

t. Connect CH A LEVEL (located on rear of counter) to oscilloscope CH A. (Do not change vertical sensitivity of oscilloscope.)

u. Adjust LEVEL CONTROL on counter for level display on the center graticule of the vertical display of the oscilloscope.

v. Repeat steps s through u, above, for CH B. (TARGET switch OFF position, CLUTTER switch ON position).

NOTE: Pulse should have sharp rise time. To sharpen leading edge of the pulse, add coaxial T's to both CH A and B of counter or oscilloscope and add resistive loading.

w. Set counter controls for time interval A to B. Set gate time for 1 nanosecond resolution. Record delay on counter display per 3.5.18.3b.

x. Convert the delay step w from nanoseconds to meters per 3.5.18.3c.

y. Record the range delay in meters as the RANGE calibration data per 3.5.18.3e.

z. Repeat steps o through y, above, for RANGE SELECT positions R2, R3, and R4.

4.7.3.17 Radar test.

a. Connect the power cables (W1, W3) and signal cables (W2, W4) as shown in Figure 3. (Signal cables are not necessary for SELF TEST procedures and are connected at this time only for convenience.) Do not connect the flexible waveguide cable W5.

b. With equipment properly interconnected, set control assembly controls as specified in Table XVII.

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TABLE XVII. Radar test settings

CONTROL	POSITION
POWER	OFF
FUNCTION SELECT	RF MONITOR
VOLTAGE SELECT	+24
RF LEVEL	20 dB
FREQ SELECT	9.229
MODE SELECT	SELF TEST
PHASE CONTROL	MID RANGE
CLUTTER	OFF
CLUTTER LEVEL	20 dB
WGE SELECT	R3
TARGET LEVEL	27 dB
TARGET FREEZE	OFF
TARGET	OFF

c. Set the Radar Set MODE SELECT switch to STANDBY.

d. Set POWER switch (located on control assembly) to ON position. Ensure that digital display (DVM) is illuminated.

e. Connect one end of flexible waveguide W5 to RADAR port (J5) and other end to SELF TEST port (J6) on front panel of the control assembly.

NOTE: Ensure pin keying on flexible waveguide cable W5 flanges is correct.

f. Set CLUTTER switch to ON position. Set CLUTTER LEVEL dial to 20.

g. Adjust RF LEVEL control to obtain indication of +2.500 fO.025 on digital display (DVM).

NOTE: Step g ensures that a calibrated power level is injected into the target and clutter channels. Therefore, monitor and maintain the adjustment delineated in step g throughout the remaining procedures unless otherwise indicated.

h. Set FUNCTION SELECT switch to RF SELF TEST position.

i. Adjust CLUTTER BALANCE for an indication on digital display (DVM) as delineated on System Tester Calibration decal.

IMPORTANT: MAINTAIN RF MONITOR LEVEL OF STEP g DURING ACCOMPLISHMENT OF STEP i.

j. Set CLUTTER LEVEL control to 25. Set CLUTTER switch to OFF position.

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k. Set TARGET switch to ON position. Set TARGET LEVEL to 27.

l. Adjust RF LEVEL control to obtain an indication of .500 Vdc to 1.000 Vdc on the digital display (DVM).

m. Set TARGET switch to OFF position. Set CLUTTER switch to ON position.

n. Observe and note digital display (DVM) indication.

o. Set CLUTTER switch to OFF position and TARGET switch to ON position.

p. Adjust TARGET BALANCE control for same digital display (DVM) indication as in step n above.

q. Alternately repeat steps m through p until digital display (DVM) indications in steps n and p are within ±0.025 Vdc.

r. Disconnect flexible waveguide W5 from SELF TEST port (J6) of the control assembly.

s. Set control assembly controls as specified in Table XVIII.

TABLE XVIII. Radar test settings for radiate power tests

CONTROL	POSITION
FUNCTION SELECT	VOLTAGE
VOLTAGE SELECT	+24
RF LEVEL	20 db
FREQ SELECT	Radar Crystal Frequency +60 MHz
MODE SELECT	RADAR
PHASE CONTROL	Mid Range
CLUTTER	OFF
CLUTTER LEVEL	70 dB
RANGE SELECT	R1
TARGET LEVEL	70 dB
TARGET FREEZE	OFF
TARGET	OFF
POWER	ON

t. Connect flexible waveguide cable W5 to Radar Set AN/VPS-2 RF OUTPUT/INPUT port. (When mounting do not cross-guide waveguide.)

u. Observe digital display (DVM) for proper power input voltage +24.00 ± 6.000 Vdc).

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v. Set VOLTAGE SELECT switch to other positions in sequence (+5, +15, -15, +28) and observe digital display (DVM) for correct output voltage indications (+5.000 ±0.200 Vdc, +15.00 ±0.300 Vdc, -15.00 ±0.300 Vdc, +28.00 ±0.600 vdc).

w. Energize Radar Set to RADIATE POWER conditions by setting SERV RAD CONT switch S6 inside top cover of Transmitter Receiver Unit 2 to ON.

x. Ensure that LOCK ON indicator is not illuminated and SIGNAL CHANNEL meter indicates less than 1.0 Vdc.

y. Set FUNCTION SELECT switch to RF MONITOR position.

z. Adjust RF LEVEL control for indication of +2.500 ± 0.025 on digital display (DVM).

NOTE: Steps y and z, above, ensure that a calibrated signal is injected into the target and clutter channels. Therefore, they should be monitored and maintained throughout the following procedures, unless otherwise indicated.

4.7.3.17.1 RF peak transmitted power measurement.

a. Set FUNCTION SELECT switch to RF INPUT position.

b. Adjust RF LEVEL Control for an indication of +2.500 ± .050 on digital display (DVM).

c. Determine the radar set RF peak transmitted power by the following calculation: RF LEVEL setting step b + RF INPUT on system tester calibration decal = radar set peak transmitted power in dBm.

4.7.3.17.2 Minimum discernible signal (MDS).

a. Set TARGET switch to ON position.

b. With FUNCTION SELECT switch in RF MONITOR position, adjust the RF LEVEL control and maintain digital display (DVM) indication of 2.500 ±0.025 Vdc for the remainder of this procedure.

c. Decrease (lower number on dial) TARGET LEVEL control until LOCK ON indictor illuminates. Note TARGET LEVEL dial number.

d. Increase (higher number on dial) TARGET LEVEL by 10.0 dB. LOCK ON light should not be illuminated after an approximate 3 second delay.

e. Repeat steps c and d, above, until repeatability occurs in step c within 1.0 dB.

f. Record TARGET LEVEL dial setting. This is the MDS of the radar set.

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4.7.3.17.3 Range accuracy measurement.

a. Set TARGET switch to ON position.

b. Adjust TARGET LEVEL control until LOCK ON indicator illuminates.

c. Decrease (lower number on dial) TARGET LEVEL control setting by 10.

d. Set FUNCTION SELECT switch to RANGE position.

e. Read and record range voltage indicated on digital display (DVM).

f. Set RANGE SELECT switch to R2, R3, and R4 positions and repeat step e, above.

NOTE: Allow 3 seconds for the Radar to re-acquire the target signal each time the RANGE SELECT switch position is changed.

g. Set FUNCTION SELECT switch to RANGE REF position.

h. Read and record range reference voltage indicated in digital display (DVM).

i. Determine range of the Radar Set by the following calculation:

$$\text{Calculated range (meters)} = \frac{\text{Range Voltage (step e),}}{\text{Range Reference Voltage (step h)}} \times 5000$$

4.7.3.17.4 Sub-clutter visibility (SCV) measurement.

a. Connect TPS short on J6 of Radar Receiver Unit 3.

b. Set RANGE SELECT to R4.

c. Set TARGET switch to ON position.

d. Adjust TARGET LEVEL control until LOCK ON indicator is illuminated.

e. Decrease (lower number on dial) TARGET LEVEL control setting by 5 dB.

f. Set TARGET FREEZE switch to ON position. (LOCK ON indicator should stay illuminated.)

g. Set CLUTTER switch to ON position.

h. Adjust CLUTTER LEVEL control until SIGNAL CHANNEL meter indicates 1 to 2 volts.

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i. Adjust PHASE CONTROL for maximum signal indication on SIGNAL CHANNEL meter.

j. Set TARGET, TARGET FREEZE, and CLUTTER switches to OFF position.

k. Set CLUTTER LEVEL **control** to 70.

l. Set CLUTTER switch to ON position.

m. With FUNCTION SELECT switch in RF MONITOR position, adjust the RF LEVEL control and maintain digital display (DVM) indication of **+2.500** ± 0.025 Vdc for the remainder of this procedure.

n. Adjust CLUTTER LEVEL control until LOCK ON indicator is illuminated. Note CLUTTER LEVEL dial number.

o. Increase (higher number on dial) CLUTTER LEVEL control setting by 10.0.

p. LOCK ON indicator should not be illuminated after an approximate 3 second delay.

q. Repeat steps n through p, above, until repeatability occurs in step n within 1.0 dB.

r. Record CLUTTER LEVEL dial reading (step n). This is the clutter power.

s. Determine SCV of the radar set by the following calculation:

SCV = MDS (4.7.3.17.2f) - Clutter Power (4.7.3.17.4r)

4.7.3.17.5 AGC threshold and linearity measurement.

a. Disconnect TPS short from J6 of Radar Receiver Unit 3.

b. With FUNCTION SELECT switch in RF MONITOR position, adjust RF LEVEL control and maintain digital display (DVM) indication of +2.500 ± 0.025 Vdc for the remainder of this procedure.

c. Set CLUTTER switch to ON position. Decrease CLUTTER LEVEL control setting slowly to 5 while observing LOCK ON Indicator. LOCK ON indicator shall not illuminate.

d. Set CLUTTER switch to OFF position.

e. Set TARGET switch to ON position.

f. Adjust TARGET LEVEL to MDS (4.7.3.17.2f) LOCK ON indicator shall illuminate.

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g. Set CLUTTER switch to ON position; LOCK ON indicator shall not illuminate.

h. Increase (higher number on dial) CLUTTER LEVEL control setting until LOCK ON indicator is illuminated. Note CLUTTER LEVEL dial reading.

i. Decrease (lower number on dial) CLUTTER LEVEL setting by 10.0. LOCK ON indicator shall not be illuminated after an approximate 3 second delay.

j. Repeat steps h and i, above, until repeatability occurs in step h within 1.0 dB.

k. Record CLUTTER LEVEL dial reading. This is the clutter power.

l. Determine the AGC threshold by the following calculation:

AGC Threshold = MDS (4.7.3.17.2f) - Clutter power (4.7.3.17.5k)

m. Decrease (lower number on dial) CLUTTER LEVEL control setting by 10 and record dial reading. The LOCK ON indicator shall not be illuminated after an approximate 3 second delay.

n. Decrease TARGET LEVEL control setting until LOCK ON indicator illuminates. Record TARGET LEVEL dial reading.

o. Repeat steps m and n in 10 dB increments. The last increment may be less than 10. (Record each increment.)

p. Determine AGC Linearity by the following calculation:

(step m) - (step n) = Incremental AGC Linearity.

4.7.3.17.6 Range rate reference (+) and range rate reference (-) measurements.

a. Set FUNCTION SELECT switch to RANGE RATE REF (+) position.

b. Record reading indicated on digital display (DVM).

c. set FUNCTION SELECT switch to RANGE RATE REF (-) position.

d. Record reading indicated on digital display (DVM).

4.7.4 Environmental Stress Screening (ESS). The control assembly shall be tested as specified in 4.7.3 after each screening procedure for compliance with 3.6.

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4.7.4.1 Temperature Cycling. This screening procedure shall be performed as specified in 3.6.1.1.

4.7.4.2 Vibration. This screening procedure shall be performed as specified in 3.6.1.2.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in paragraph 3.8.

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet PS11748205. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The control assembly consists of microwave and digital components necessary to provide RF and digital control and measurement functions required to accurately determine the radar operational status of the AN/VPS-2 Vulcan Radar set.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Applicable stock number.
c. First Article Sample requirements.
d. Packaging requirements, if other than specified in section 5.
e. Serialization requirements, if applicable.
f. Certificate of Conformance for each lot or shipment of product.
g. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF-1, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

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6.5 Subject term (key word) listing.

System Tester, Radar, ANIVPM-2
Vulcan Radar Set, AN/VPS-2

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the pxevious issue due to the extensiveness of the changes.

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Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-48605A (AR)	14 June 1991

3. DOCUMENT TITLE: CONTROL ASSEMBLY R.F. (UNIT 1) 11748205

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON (*If applicable*)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (*Include Area Code*)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (*Include Zip Code*) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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INCH-POUND

MIL-C-48605A (AR)
AMENDMENT 1
25 September 1992

MILITARY SPECIFICATION

CONTROL ASSEMBLY R.F. (Unit 1): 11748205

This Amendment forms a part of Military Specification MIL-C-48605A, dated, 14 June 1991, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1.1: Add the following military specification:

"MIL-T-28800 - Test Equipment for use with Electrical and Electronic Equipment, General Specification for"

PAGE 2

2.1.1 Delete the following military standard:

"MIL-STD-2000 - Standard Requirements for Soldering Electrical and Electronic Assemblies"

2.1.2: Add the following drawing:

"12950877 - Requirements for Soldering Electrical and Electronic Assemblies"

PAGE 3

3.3.1: Delete paragraph in its entirety and substitute:

"3.3.1 Vibration. The control assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as specified in MIL-T-28800 for Type 1, Class 2, Style A test equipment."

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PAGE 3

3.3.2, 3.3.2.1 and 3.3.2.2: Delete paragraphs in their entirety and substitute:

"3.3.2 Altitude Operating. The control assembly shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg)."

PAGE 18

3.8: Delete paragraph 3.8 in its entirety and substitute:

"3.8 Soldering. All soldering shall be in accordance with the requirements of Note 2 of DRawing 12950877."

PAGE 23

4.7.1.1: Delete paragraph 4.7.1.1 in its entirety and substitute:

"4.7.1.1 Vibration. This test shall be performed as specified in MIL-T-28800 for Type I, Class 2 Style A test equipment with the exception that Step 4.c Resonance Dwell does not apply. The control assembly shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the control assembly sahll be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1."

PAGE 24

4.7.1.2: Delete paragraph 4.7.1.2 in its entirety and substitute:

"4.7.1.2 Altitude Operating. This test shall be performed as specified in Procedure II, Method 500, of MIL-STD-0810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.2."

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PAGE 44

4.8: Delete paragraph 4.8 in its entirety and substitute:

"4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of Note 2 of Drawing 12950877 as specified in paragraph 3.8."

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1240-AB37)

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